



Fedja Bogataj

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EDUCATION

- **Technical University of Vienna (TU Wien)** Vienna, Austria
Master of Science in Data Science; GPA: 1.23 Mar. 2025 – Present
 - **Specializations:** Machine Learning and Statistics, Big Data and High Performance Computing
 - **Relevant Coursework:** Machine Learning, Advanced Methods for Regression and Classification, Statistical Simulation and Computer-intensive Methods, AKSTA Statistical Computing, Experiment Design and Execution
- **University of Ljubljana, Faculty of Computer and Information Science** Ljubljana, Slovenia
Bachelor of Computer Science; Avg. grade: 9.00/10 Oct. 2021 – Sep. 2024
 - **Specialization:** Machine Learning and Artificial Intelligence
 - **Relevant Coursework:** Probability and Statistics, Mathematical Modelling, Economics and Entrepreneurship (business decision-making, accounting statements, time value of money)

PROJECTS

- **Deep Reinforcement Learning for Market Prediction (Bachelor's Thesis):** Designed and compared feedforward, LSTM, and GRU-based Deep Q-Networks in **Python** to predict buy/hold/sell decisions directly from historical price data across multiple financial instruments (2019 – 2024), with the GRU variant a largely unexplored architecture for this domain. Adapted and tuned the reward function and state representation so agent behavior reflected the underlying value structure in the data, rather than fitting to surface-level price movement. Evaluated and compared all three architectures on the same instruments to isolate how much of the performance difference came from the recurrent structure itself versus the trading strategy learned.
- **Industrial AI – Process Logic Benchmarking (Zero One Hack 2026):** Built and trained transformer models in **Python** for sequence prediction and anomaly detection on time-ordered process data; trained and scheduled via SLURM on the Leonardo EuroHPC cluster; evaluated with Top-k accuracy, MRR, F1, and ROC-AUC using **scikit-learn**.
- **Teapot – AI Quality Control for Fiber-Optic Trench Documentation (ViennaUp Hackathon 2026):** Built a computer-vision pipeline in **Python** that screens construction-site images for compliance and quality-control anomalies, collaborated in a small team under hackathon time pressure and presented findings to judges.
- **Vowi 2.0:** Built a Retrieval-Augmented Generation application in **Python** using ChromaDB as the vector database, developed during Master's studies.
- **Notepilot:** A Retrieval-Augmented Generation (RAG) based note-taking assistant application, built in **Python**.

EXPERIENCE

- **SRC d.o.o.** Ljubljana, Slovenia
Full Stack Developer Oct 2024 – Mar 2025
 - **Delivery:** Led development of a **.NET / C#** microservices platform for internal and cross-enterprise document sharing and e-signing, with a Swagger/OpenAPI setup.
 - **Architecture & Requirements:** Ran requirements-gathering sessions directly with clients, then designed the system architecture and database end-to-end from those requirements.
 - **Mentoring:** Mentored an intern on frontend development.
- **SRC d.o.o.** Ljubljana, Slovenia
Software Engineer - Part time Jul 2023 – Oct 2024
 - **Microservices Integration:** Built core features of a **.NET/Vue.js** visitor management microservice, integrating third-party **APIs** for QR code scanning and partner business data.

- **Collaboration:** Held meetings and product demos with an external hardware partner, presenting progress and collaborating on integrating their hardware and software into the platform.
- **Deployment:** Deployed the application on **Microsoft Azure**.

- **SRC d.o.o.**
Application Tester

Ljubljana, Slovenia
Aug 2022

- **QA:** Performed structured application testing using SoapUI, covering functional and API-level test cases.

SKILLS

- **Programming:**
 - **Proficient:** Python, SQL, C#, R
 - **Familiar:** C, C++, Java
- **Quantitative & ML:** pandas, scikit-learn, TensorFlow, Reinforcement Learning (DQN, LSTM, GRU), Transformers, statistical modeling, time-series data
- **Data & Tools:** PostgreSQL, ChromaDB, Git, GitLab, Jenkins, Docker, Azure, SLURM/HPC
- **AI-Assisted Tools & Continuous Learning:** Claude Code, Cursor, active Kaggle competitor, self-directed learning of new tools and frameworks

LANGUAGES

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|-----------------------------------|--------------------|-------------------|
| • Slovenian: Mother tongue | English: C1 | German: B1 |
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EXTRACURRICULAR ACTIVITIES

- **Board of European Students of Technology (BEST) Vienna:** Full member since December 2025; logistics responsible for beWANTED 2026 career fair and Summer Course 2026.
- **Competitive Swimming:** National-level competitor in Slovenia (2008 – 2021), training 5–10 times per week; gold and bronze medals at international competitions.